

# The Automated FDE — Roles, Responsibilities & Lifecycle

*An 80:20 (AI : Human-in-the-Loop) split for Forward-Deployed Engineering*

**The goal:** automate the Forward Deployed Engineer's lifecycle so that ~80% of the mechanical work is executed by an agentic AI FDE (the swarm), and ~20% is reserved for the human FDE — the Pod Commander — who supplies judgment, trust, architecture, and governance. This document defines the role on both sides of that line and the lifecycle that connects them.

**The 80:20 is a leverage ratio, not a uniform rule.** Some stages are 95% AI (boilerplate codegen); others are 80% human (architecture sign-off, client trust, regulated-data governance). The discipline is putting the human 20% exactly where judgment is irreplaceable — and letting the AI own everything else.

THE HEADLINE SPLIT



**THE GOVERNING PRINCIPLE**

AI provides the human decision. The agentic FDE generates, tests, and self-heals at machine speed. The human FDE owns the prerogative — the architectural call, the trust in the team, the compliance gate. The 80% buys velocity, the 20% buys safety.

“FDE” is now a two-part entity: the **Agentic FDE** (the 80%) and the **Human FDE / Pod Commander** (the 20%). They are not peers doing the same job — they own different classes of work, divided by what only a human can be accountable for.

### The Agentic FDE — the 80% (machine-owned)

A multi-agent swarm of specialized roles, each a node in the lifecycle. It owns **velocity, breadth, and repetition**.

| Agent role                | Owns                                                                           | Lifecycle stage |
|---------------------------|--------------------------------------------------------------------------------|-----------------|
| <b>Spec Architect</b>     | Translates the human's intent + spec into a rigorous, machine-executable plan. | 1–2             |
| <b>Systems Engineer</b>   | Writes the code, resolves conflicts, self-corrects syntax, builds the binary.  | 3               |
| <b>Adversarial SDET</b>   | Generates and runs unit/integration/E2E tests; stress-tests its own output.    | 4               |
| <b>Security Sentinel</b>  | Runs SAST, dependency scans, and compliance checks as agent tools.             | 5               |
| <b>DevOps Overseer</b>    | Manages PRs, canary deploys, rolling updates within guardrails.                | 6               |
| <b>Reliability Agent</b>  | Monitors production, maps stack traces, cuts auto-hotfix PRs.                  | 7               |
| <b>Platform Architect</b> | Abstracts reusable components from the Spoke back up to the Hub (background).  | cross-cutting   |

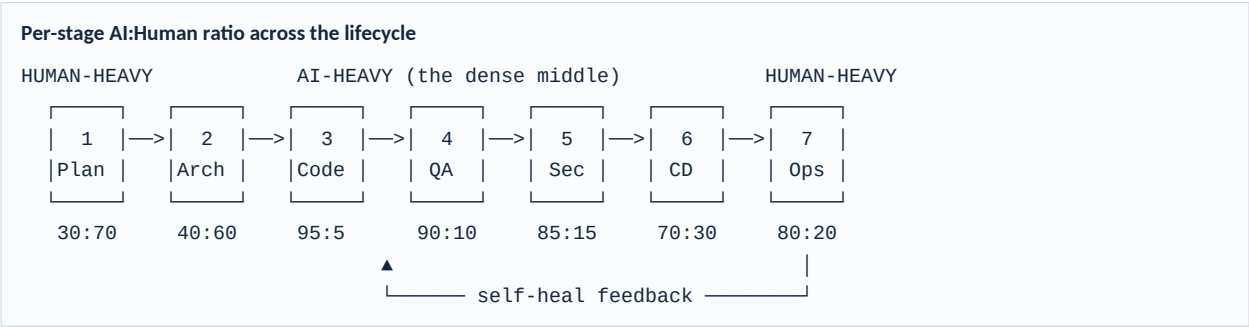
### The Human FDE / Pod Commander — the 20% (judgment-owned)

One senior, embedded, customer-trusted engineer. They do not out-type the swarm; they *out-judge* it. They own **context, trust, architecture, and governance** — the four things an agent cannot be accountable for.

| Human responsibility                  | Why it can't be automated                                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Diagnosis &amp; architecture</b>   | Reading the client's real problem — the gap that “lives between the tools” — requires judgment the swarm has no context for. |
| <b>Client trust &amp; evangelism</b>  | A CISO trusts a person in the room, not a PR bot. Change leadership is human work.                                           |
| <b>Spec authorship &amp; intent</b>   | The swarm builds what the spec says; a wrong spec fails at machine speed. The human owns intent.                             |
| <b>Governance &amp; the ship gate</b> | In regulated domains, a human is accountable for the irreversible call. The gate is never delegated.                         |
| <b>Exception triage</b>               | When an agent loops or hits an edge case, the human breaks the tie and re-plans.                                             |

PART II • THE 80:20 FDE LIFECYCLE, STAGE BY STAGE

The split is per-stage. Below, each of the seven stages shows its AI:Human ratio, what the swarm does, what the human does, and the gate that separates them. Note the shape: the human owns the **front** (intent + architecture) and the **end** (the ship gate) — the swarm owns the dense middle.



Stage 1 — Context Discovery & Planning • AI 30% / HITL 70%



**Agentic FDE (AI):** Ingests the codebase topology, parses Jira/Linear issues, retrieves prior art from the vector graph, drafts a candidate execution plan.

**Human FDE (HITL):** Sets the intent and priorities, frames the real problem, validates the plan against client reality, approves scope.

**Gate:** Human approves the plan before any code is touched.

Stage 2 — Architecture & Design • AI 40% / HITL 60%



**Agentic FDE (AI):** Queries documentation, proposes API specs, checks schema compatibility, surfaces design options with trade-offs.

**Human FDE (HITL):** Makes the architecture call, enforces governance constraints, accepts/rejects the design. The irreversible decisions live here.

**Gate:** Human signs off on the architecture and the contracts (ADRs frozen).

Stage 3 — Implementation & Coding • AI 95% / HITL 5%



**Agentic FDE (AI):** Multi-agent coding squads write the code, resolve merge conflicts, self-correct syntax, build the binary on the client's stack.

**Human FDE (HITL):** Spot-reviews novel logic; otherwise stays out of the way. This is the swarm's stage.

**Gate:** No gate mid-stage; output flows into automated QA.

Stage 4 — Agentic Verification (QA) • AI 90% / HITL 10%



**Agentic FDE (AI):** Generates unit/integration/E2E tests from the acceptance criteria; the Adversarial SDET stress-tests and kicks failures back to coding.

**Human FDE (HITL):** Reviews coverage on high-risk paths; confirms the acceptance criteria were the right ones.

**Gate:** Automated: 100% test pass + coverage threshold before security.

**Stage 5 — Security & Guardrails** • AI 85% / HITL 15%



**Agentic FDE (AI):** Runs SAST, dependency and license scans, compliance checks; blocks on severity-1 findings.

**Human FDE (HITL):** Owns the compliance interpretation in regulated domains; signs the risk acceptance where required.

**Gate:** Human signs regulated-data / compliance acceptance; else automated.

**Stage 6 — Continuous Deployment** • AI 70% / HITL 30%



**Agentic FDE (AI):** Opens the PR, runs canary and rolling deploys within branch-protection guardrails, watches health.

**Human FDE (HITL):** Holds the ship gate. In Embedded mode the human approves the merge; in Remote mode the automated gate does, under human-set policy.

**Gate:** THE SHIP GATE — human approval (Embedded) or policy gate (Remote).

**Stage 7 — Observability & Self-Healing** • AI 80% / HITL 20%



**Agentic FDE (AI):** Monitors production, detects anomalies, maps stack traces, auto-cuts hotfix PRs, re-enters the loop at stage 1.

**Human FDE (HITL):** Triage escalations the swarm can't resolve; tunes the guardrails and budgets; owns the post-incident call.

**Gate:** Auto-heals within policy; escalates to human on repeated failure.

Who is Responsible, Accountable, Consulted, Informed across the lifecycle. The pattern that matters: the **human is Accountable for every stage** even where the AI is Responsible for the work. Accountability never automates.

| Stage            | Agentic FDE (AI)      | Human FDE                   | Accountable |
|------------------|-----------------------|-----------------------------|-------------|
| 1. Plan          | Responsible (drafts)  | Approves intent/scope       | Human       |
| 2. Architecture  | Consulted (proposes)  | Responsible (decides)       | Human       |
| 3. Code          | Responsible (builds)  | Informed / spot-review      | Human       |
| 4. QA            | Responsible (tests)   | Consulted (coverage)        | Human       |
| 5. Security      | Responsible (scans)   | Accountable (risk sign-off) | Human       |
| 6. CD            | Responsible (deploys) | Accountable (ship gate)     | Human       |
| 7. Observability | Responsible (heals)   | Consulted (escalations)     | Human       |

### The leverage math — why 80:20 holds

One human FDE supervises a swarm that executes the equivalent of a multi-engineer team. The ratio is not about effort hours — it is about **where accountability and judgment must sit**. The 80% is delegated because it is mechanical and verifiable; the 20% is retained because it is irreversible or unquantifiable.

| What scales (the 80%)                     | What doesn't (the 20%)                             |
|-------------------------------------------|----------------------------------------------------|
| Code generation, refactoring, glue code   | The architecture decision and its trade-offs       |
| Test authoring and execution              | Whether the acceptance criteria are even correct   |
| Security scanning and dependency hygiene  | The compliance interpretation and risk sign-off    |
| Canary deploys, rollbacks, hotfix PRs     | The “ship / don't ship” call in a regulated domain |
| Production monitoring and anomaly mapping | Client trust, evangelism, and change leadership    |

The 80:20 ratio shifts with topology. Embedded PODs keep the human closer to the loop (more HITL); Remote PODs push toward autonomy (less HITL, but a stricter automated gate). The human's **accountability is constant**; only the interaction surface changes.

| Dimension       | Embedded POD                               | Remote POD                              |
|-----------------|--------------------------------------------|-----------------------------------------|
| Effective split | ~70:30 (AI:Human) — human reviews more     | ~90:10 (AI:Human) — automated gates     |
| The ship gate   | Human approves the merge in Slack/GitHub   | Automated policy gate, human-defined    |
| Best for        | Legacy apps, core features, regulated work | Greenfield, ephemeral micro-services    |
| Trust posture   | “Super-powered intern” earning trust       | Sovereign service, trust already earned |
| Human role      | Reviewer + architect + evangelist          | Policy-setter + exception handler       |

**The adoption path:** start Embedded (70:30) to earn trust on the client's PR outputs, then offer the toggle to Remote (90:10). Trust is earned in Embedded and spent in Remote — the same land-and-expand logic that governs the whole FDE practice.

### The guardrails that make the 80% safe

1. **Branch protection.** Agents never push to main/production — every change is a PR. The human (or policy) holds merge.
2. **Token & compute budgets.** Hard per-task spend caps; breach triggers teardown + SRE alert. The kill-switch is non-negotiable.
3. **Loop detection.** If an agent fails the same test >5 times, it stops and escalates to the human triage queue.
4. **Deterministic gates.** 100% test pass + coverage threshold + zero sev-1 security before any deploy. Non-negotiable, automated.
5. **Unified telemetry.** Every agent action is logged to a central observability layer — the human supervises the swarm, not each keystroke.

#### THE 80:20 IN ONE LINE

Automate the 80% that is mechanical, verifiable, and reversible. Retain the 20% that is judgment-bound and irreversible. The agentic FDE gives the velocity of a team; the human FDE gives the accountability of a senior engineer. AI proposes; the human disposes.